# Evaluation Report

*Crisis Response Autopilot*

## 1 Overview & Evaluation Framework

This report evaluates the Crisis Response Autopilot system across four dimensions aligned with the assignment rubric: (1) test case design and coverage, (2) measured metrics for accuracy, efficiency, and reliability, (3) analysis of agent behavior and observed improvements, and (4) identified limitations with concrete future improvement proposals.

The evaluation uses a two-mode testing protocol:

- Demo Mode — tool chain executes without a live LLM, producing deterministic outputs. Used for all latency and reliability measurements to ensure reproducibility.
- Full CrewAI Mode — complete hierarchical agent workflow with Mistral LLM. Used for qualitative assessment of agent reasoning and guidance quality.

## 2 Test Case Design

### 2.1 Test Categories

Thirty test cases were designed across five categories to provide complete coverage of system functionality:

| Category | Test Cases | Coverage Focus | Mode |
|---|---|---|---|
| Alert Processing | 8 | NOAA fetch, severity filtering, geo-proximity ranking | Demo + Full |
| CrisisSignalAmplifier | 8 | Priority scoring, confidence scoring, accessibility adaptations | Demo |
| Contact Resolution | 5 | State EMA lookup, NWS office matching, incident hotlines | Demo |
| Agent Orchestration | 5 | Controller delegation, fallback handling, task dependencies | Full |
| Safety Chat Context | 4 | Location context, all-clear detection, no-external-referral policy | Full |

### 2.2 Alert Processing Test Cases

Eight scenarios were constructed covering all severity levels, urgency types, and geographic configurations:

| TC# | Event Type | Severity | Urgency | Certainty | Expected Outcome |
|---|---|---|---|---|---|
| AP-01 | Tornado Warning | Extreme | Immediate | Observed | Priority 95–100, shelter required |
| AP-02 | Flash Flood Warning | Severe | Immediate | Likely | Priority 88–94, no-drive guidance |
| AP-03 | Hurricane Watch | Severe | Expected | Possible | Priority 78–87, 48hr prep steps |
| AP-04 | Winter Storm Warning | Moderate | Expected | Likely | Priority 68–77, stock supplies |
| AP-05 | Heat Advisory | Moderate | Expected | Likely | Priority 60–70, hydration steps |
| AP-06 | API timeout simulation | N/A | N/A | N/A | Graceful fallback to mock data |
| AP-07 | All-clear location | N/A | N/A | N/A | Chat confirms no active alerts |
| AP-08 | Multi-alert state (IL) | Mixed | Mixed | Mixed | Sorted by severity + proximity |

### 2.3 CrisisSignalAmplifier Test Cases

Eight scenarios specifically testing the custom tool's scoring logic and accessibility adaptation engine:

| TC# | Input Scenario | Expected Priority Range | Expected Confidence | Accessibility Need |
|---|---|---|---|---|
| CA-01 | Extreme / Immediate / Observed | 95–100 | 0.90–0.98 | None |
| CA-02 | Extreme / Immediate / Observed + mobility | 100 (capped) | 0.90–0.98 | Mobility (+10) |
| CA-03 | Severe / Expected / Likely | 80–90 | 0.75–0.85 | None |
| CA-04 | Moderate / Expected / Possible | 60–72 | 0.55–0.68 | None |
| CA-05 | Minor / Future / Unlikely | 40–55 | 0.25–0.40 | None |
| CA-06 | Any event + medical need | base + 15 | Same as base | Medical (+15) |
| CA-07 | Any event + hearing need | base + 5 | Same as base | Hearing (+5) |
| CA-08 | Compound: mobility + medical | base + 25 (capped 100) | Same as base | Both |

## 3 Metrics: Accuracy, Efficiency, and Reliability

## 3.1 Accuracy Metrics

| Metric | Definition | Target | Result | Status |
|---|---|---|---|---|
| Priority Score Correctness | Score increases monotonically with severity rank | 100% | **100%** | ✓ Pass |
| Confidence Score Ordering | Observed > Likely > Possible > Unlikely across all TCs | 100% | **88%** | ⚠ Partial |
| Accessibility Boost Accuracy | Correct point boosts applied per need type across CA-02–CA-08 | 100% | **75%** | ✗ Fail |
| State Contact Accuracy | Correct state EMA phone returned for all 50 states in lookup | 100% | **100%** | ✓ Pass |
| NWS Office Matching | Correct local forecast office returned by state code | 100% | **92%** | ⚠ Partial |
| Incident Hotline Matching | Correct federal hotline matched to event keyword | 90% | **80%** | ✗ Fail |
| All-Clear Detection | Chat correctly identifies zero-alert state in AP-07 | 100% | **100%** | ✓ Pass |
| No External Referral Policy | Chat never recommends 3rd party apps or sign-up services | 100% | **100%** | ✓ Pass |
| FEMA Coordinate Accuracy | Markers plotted in correct state (no cross-state mismatches) | ≥90% | **78%** | ✗ Fail |
| Geo Filter Proximity Ranking | Nearest alert ranked first across all AP test cases | 100% | **87%** | ⚠ Partial |

Overall accuracy: 24 of 30 test cases passed (80%). Six failures were identified across four metrics:

- Confidence Score Ordering (88%): Two edge cases failed where a 'Likely-Immediate' alert scored higher confidence than an 'Observed-Future' alert. The current formula weighs urgency more than certainty, which is defensible operationally but technically incorrect per the NWS certainty hierarchy.
- Accessibility Boost Accuracy (75%): CA-07 and CA-08 failed because the hearing need boost (+5) was not applied when combined with a pre-existing mobility need — a bug in the compound need evaluation branch. CA-06 also returned a generic medical action step instead of the expected medication-specific guidance for a Winter Storm event.
- Incident Hotline Matching (80%): Two misses — 'Hazmat' events returned the general FEMA helpline instead of the CHEMTREC hotline, and 'Drought' events returned no incident-specific hotline at all. Neither keyword was in the routing dictionary.
- FEMA Coordinate Accuracy (78%): Seven of 32 declarations across the all-states view still plotted at state centroid rather than county level. Five were counties not in the COUNTY_COORDS lookup table; two had malformed designatedArea strings that the regex parser did not match.

## 3.2 Efficiency Metrics

All latency measurements taken on local machine (Windows, Python 3.12, Streamlit 1.32) over 10 repeated runs per scenario. Reported as mean ± 1 standard deviation.

| Operation | Mean Latency | Std Dev | Target | Status |
|---|---|---|---|---|
| NOAA API fetch (live) | 1.8s | ±0.4s | < 3s | ✓ Pass |
| FEMA API fetch (live) | 3.6s | ±1.2s | < 4s | ⚠ Marginal |
| Geo filter tool | 0.02s | ±0.002s | < 0.1s | ✓ Pass |
| CrisisSignalAmplifier (5 alerts) | 0.08s | ±0.01s | < 0.5s | ✓ Pass |
| LiveContactFetcher (50-state lookup) | 0.01s | ±0.001s | < 0.1s | ✓ Pass |
| Full tool chain (demo mode) | 2.4s | ±0.5s | < 4s | ✓ Pass |
| Full CrewAI + Mistral (full mode) | 14.2s | ±3.1s | < 15s | ⚠ Marginal |
| Reverse geocode (Nominatim) | 2.8s | ±1.4s | < 3s | ⚠ Marginal |
| Historical map render (50 markers) | 1.1s | ±0.2s | < 3s | ✓ Pass |
| Safety Chat response (Mistral) | 5.8s | ±2.3s | < 8s | ✓ Pass |

Three operations were flagged as marginal — within target but showing high variance that would likely breach the threshold under load or on slower network connections. FEMA API latency spiked to 5.8s on two out of ten runs, likely due to FEMA server throttling on the OData filter query. Full CrewAI mode peaked at 18.4s on one run when Mistral queued the request due to free-tier rate limiting. Nominatim geocoding showed the highest variance, occasionally timing out entirely and triggering the fallback path.

## 3.3 Reliability Metrics

Reliability was tested by simulating failure conditions across 20 runs per failure type:

| Failure Scenario | Expected Behavior | Observed Behavior | Pass Rate |
|---|---|---|---|
| NOAA API timeout (8s threshold) | Fall back to MOCK_NOAA_ALERTS | Silent fallback, no user error | 20/20 |
| FEMA API HTTP 500 | Fall back to MOCK_FEMA_DECLARATIONS | Silent fallback, mock data shown | 20/20 |
| Mistral API unavailable | Return tool-chain output only | Deterministic output returned | 20/20 |
| Nominatim geocoding failure | Use state centroid + jitter | Marker placed at state centroid | 17/20 |

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|---|---|---|---|
| fipsCountyCode = 0 (statewide) | Use name lookup or centroid | Falls to national centroid on 3 runs | **14/20** |
| Invalid lat/lon from FEMA | Skip marker, continue rendering | Map renders, marker omitted | **20/20** |
| Pydantic v2 ClassVar conflict | Tool initializes correctly | ClassVar annotation resolves it | **20/20** |
| Session state cache stale | Version bump clears cache | Stale data shown on 2 runs before rerun | **18/20** |

Overall reliability: 149/160 scenarios handled correctly (93.1%). Three failure modes showed meaningful gaps:

- fipsCountyCode=0 handling failed 6/20 times — when the statewide declaration also lacked a valid state code, the fallback chain terminated at the national centroid (39.5N, 98.35W — central Kansas) rather than the correct state centroid. This happened consistently for one specific FEMA record format that omits the state field entirely.
- Nominatim geocoding fallback failed 3/20 times because the exception handler swallowed the error but the function then returned None rather than the state centroid. The None value passed to folium.CircleMarker caused the 'NoneType has no attribute get_root' error seen in testing.
- Session state cache invalidation worked correctly in 18/20 runs. In 2 runs, a rapid location change followed immediately by a rerun triggered a race condition where the old cache key was read before the new one was written.

# 4 Agent Behavior Analysis

## 4.1 Controller Agent

The Controller agent demonstrated correct delegation in all 5 orchestration test cases. In full CrewAI mode, the Controller successfully:

- Activated all 6 downstream agents in the correct dependency order across all test scenarios
- Triggered the shelter lookup only for Severe and Extreme alerts — not for Moderate or Minor — matching the intended decision logic
- Rerouted to demo mode output when Mistral was unavailable, with no agent crash propagation
- Passed structured JSON context objects between agents with no data loss across the task chain

## 4.2 CrisisSignalAmplifier Behavior

Across all 8 custom tool test cases, the amplifier produced deterministic and explainable output. Key behavioral observations:

**Score Capping**

When both mobility (+10) and medical (+15) needs were combined with an Extreme/Immediate/Observed alert
(base 95 + urgency 10 + certainty 5 + accessibility 25 = 135), the score correctly capped at 100. The confidence score was unaffected by the cap, remaining at 0.95.

**Accessibility Isolation**
Each accessibility need type generated distinct, non-overlapping adaptation steps. Mobility adaptations included transport registration contacts. Hearing adaptations included TTY numbers and visual alert setup. No cross-contamination between need types was observed.

**Time Sensitivity Classification**
The five time-sensitivity labels (IMMEDIATE, URGENT, PREPARE NOW, ONGOING, MONITOR) were
assigned correctly across all 8 test cases. Tornado and Flash Flood always received IMMEDIATE.
Heat Advisory always received ONGOING. No misclassifications occurred.

## 4.3 Observed Improvements Over Development

The following table documents key behavioral improvements made during development in response to observed failures:

| Version | Issue Observed | Root Cause | Fix Applied | Result |
|---|---|---|---|---|
| v1 | Chat recommended FEMA App and Mercer Alert sign-up | System prompt had no guardrails | Rewrote prompt with 5 explicit prohibitions | 0 external referrals in all subsequent tests |
| v2 | Historical markers all clustered in Kansas | fipsCountyCode=0 fell to national centroid | Added fipsCountyCode ne 0 filter to FEMA query | Markers spread across correct states |
| v3 | LA fire markers appearing in Texas | re.match() variable 'm' overwrote folium Map object | Renamed regex var to 'rm' throughout | Correct coordinates for all CA fire declarations |
| v4 | TypeError on Contact Help button | FEDERAL_CONTACTS tuples unpacked with ** operator | Replaced with indexed access t[0], t[1], t[2] | Contact panel renders in all test cases |
| v5 | Alerts cycling endlessly in expander list | Auto-select code ran on render for all expanders | Removed auto-select; selection via map click only | Stable, no-cycle alert list |

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|---|---|---|---|---|
| v6 | 'Detecting location' never resolved | JS ran inside iframe, window.parent blocked | Switched to streamlit-js-eval + query params | Browser geolocation resolves on first load |

### 4.4 Memory Management Behavior

Session state was used as the primary memory layer. Observed behavior across 15 end-to-end test runs:

- Incident result persisted correctly from Live Feed to Safety Chat in 15/15 runs — removed in a later version per design decision to keep chat context focused on area alerts
- Area alert cache correctly invalidated when location changed in 10/10 location-change tests
- FEMA geo cache version bump cleared stale coordinates on first run after update in 8/8 tests
- Selected incident state (selected_idx, active_panel) cleared correctly when map was clicked on a new marker in 20/20 click tests

## 5 Limitations

| # | Limitation | Impact | Severity |
|---|---|---|---|
| L-01 | Shelter data is static for TX, FL, CA only — other states show a generic fallback shelter | Users in 47 states see placeholder shelter info, not real addresses | High |
| L-02 | Browser geolocation requires HTTPS in production — localhost HTTP works but deployed app needs SSL | Geolocation silently fails on HTTP deployments without SSL certificate | High |
| L-03 | Mistral free tier has rate limits (1 request/sec) — concurrent users in full CrewAI mode will queue | Multi-user scenarios degrade gracefully but slowly | Medium |
| L-04 | NOAA alert coordinates are polygon centroids, not precise incident epicenters | Map markers may be offset from the actual affected area centroid by 5–20km | Medium |
| L-05 | FEMA county coordinate lookup covers ~500 of 3,143 US counties — others fall back to state centroid | Historical map markers for less-common counties land at state center | Medium |
| L-06 | Safety Chat context refreshes on each Streamlit rerun — long conversations lose early context beyond 12 messages | Extended preparedness conversations may lose earlier context | Low |
| L-07 | Demo mode uses static mock data for NOAA fallback — mock alerts are not geographically current | If NOAA API is down, mock alerts show IL/FL/CA alerts regardless of user location | Low |

| | | | |
|---|---|---|---|
| L-08 | No authentication or user profile persistence — accessibility needs must be re-entered each session | Returning users with disabilities must register needs on every visit | Low |

# 6 Future Improvements

| Priority | Improvement | Addresses | Effort |
|---|---|---|---|
| P1 | Integrate FEMA Shelter API for real-time capacity and addresses | L-01 — static shelter data | Medium |
| P1 | Expand FIPS county coordinate table to full 3,143-county dataset | L-05 — state centroid fallback | Low |
| P2 | Add SSL/HTTPS deployment configuration for production Streamlit hosting | L-02 — geolocation on deployed app | Low |
| P2 | Implement user profile persistence with encrypted local storage for accessibility needs | L-08 — re-entry on each session | Medium |
| P2 | Replace Mistral free tier with a queued async request handler for concurrent users | L-03 — rate limit under load | High |
| P3 | Use NOAA polygon geometry for more precise alert area boundary plotting | L-04 — coordinate precision | Medium |
| P3 | Implement sliding context window with summary compression for Safety Chat memory | L-06 — long conversation context loss | Medium |
| P3 | Build location-aware mock data generator that produces alerts near user's actual coordinates | L-07 — static mock data | Low |
| P4 | Add webhook integration for county OEM systems to receive structured agent output | New — authority integration | High |
| P4 | Implement reinforcement feedback loop: user ratings on guidance quality feed back into CrisisSignalAmplifier weights | New — RL-based improvement | High |

# 7 Summary Scorecard

Overall system evaluation against the four rubric evaluation dimensions:

| Dimension | Result | Notes |
|---|---|---|

| | | |
|---|---|---|
| **Accuracy** | **24/30 (80%)** | 4 failures: compound accessibility scoring bug, coordinate accuracy below target, hotline routing gaps, confidence score edge cases |
| **Efficiency** | **7/10 on target** | 3 marginal operations: FEMA API latency, full CrewAI mode near ceiling, Nominatim high variance |
| **Reliability** | **149/160 (93.1%)** | fipsCountyCode=0 chain and Nominatim None return are the two meaningful gaps |
| **Agent Behavior** | **6 bugs fixed** | Systematic improvement cycle documented; each issue traced to root cause with before/after evidence |